

Informal Worker Income Verification for Micro-Loan Eligibility:

An Explainable ML Framework Using UPI Transaction

and Mobile Recharge Behaviour

[SYNTHETIC DATA SIMULATION STUDY]

⚠ ALL DATA IN THIS REPORT IS SYNTHETICALLY GENERATED ⚠

Generated: September 29, 2026

Seed: 42 | 10,000 synthetic workers | 90-day window

Framework: XGBoost + SHAP | UPI + Mobile Recharge features

Purpose: Methodological feasibility study for research paper

Section 1

Experimental Setup and Data-Generation Description

1.1 OVERVIEW

This is a calibrated synthetic-data simulation study. No real data was used.

All results indicate methodological feasibility only and must NOT be interpreted as real-world performance. The word 'synthetic' appears in every table and figure caption involving simulated data.

1.2 DATA GENERATION PARAMETERS

- Total workers: 10,000 (5,000 repaid / 5,000 defaulted — balanced experiment)
- Secondary experiment: 5,000 workers, 80/20 imbalance (20% default)
- Worker categories: street vendor (30%), daily-wage labourer (30%), gig/delivery worker (25%), domestic worker (15%)
- Observation window: 90 days (2024-01-01 to 2024-04-01)
- Random seed: 42 (everything reproducible)

1.3 LATENT REPAYMENT CAPACITY

Each worker is assigned a latent repayment capacity $\in [0,1]$ drawn from Beta(2,2) mixed with normalised income and digital-adoption rate. This unobserved variable drives both the volume/regularity of UPI activity and the probability of default. About 15% of workers experience random income shocks that reduce capacity. Label noise of ~6% is applied (labels flipped) so the task is not trivially separable.

1.4 UPI EVENT GENERATION

- Amounts: log-normally distributed, INR 20–5,000
- Weekly pattern: higher Mon–Sat, lower Sunday
- Festival bumps: days 15–16, 45–46, 72–73 ($1.5\times$ – $2.5\times$ amount)
- Sparse workers: those with low digital-adoption $\times$ capacity can have 0–7 events
- Credit probability: $0.45 + 0.2 \times \text{capacity}$
- Counterparty pool: 3–15 unique per worker

Section 2

Dataset Summary (Synthetic)

Table 1 — Dataset Summary (Synthetic data — methodological feasibility only)

Parameter	Value
Total workers	10,000
Label=0 (repaid)	5,000
Label=1 (defaulted)	5,000
Label noise applied	~6% (random flip)
Observation window	90 days
Train size	8000
Test size	2000
Train positive rate	0.501
Test positive rate	0.501
UPI events (approx)	~500,000
Recharge events (approx)	~50,000
Total features	24
CV strategy	Stratified 5-fold inside training set
XGBoost search	RandomizedSearchCV, n_iter=20
Seed	42

Section 3

Model Performance Metrics

Table 2 — Main Model Metrics on Held-out Test Set (Synthetic data)

model	AUC	Accuracy	Precision	Recall	F1	Brier
XGBoost	0.7214	0.6525	0.6461	0.679	0.6621	0.2072
LogisticRegression	0.6977	0.628	0.6223	0.657	0.6392	0.2162
RandomForest	0.6993	0.635	0.6218	0.6949	0.6563	0.2198

Table 3 — Calibration: Brier Score Before/After Isotonic Regression (Synthetic data)

Model	Brier (before cal.)	Brier (after isotonic)	Improvement
XGBoost	0.20724889636039734	0.2023797333240509	0.0049

Table 4 — Tier Analysis: Default Rate per Score Tier — ✓ Monotonic (Synthetic data)

tier	count	default_rate	mean_score
Eligible	236	0.1483	84.68
Review	1362	0.4670	51.65
High_risk	402	0.8259	23.58

Table 5 — Tier Classification: Eligible vs High-Risk (Synthetic data)

Metric	Value
Eligible_vs_HighRisk_Accuracy	0.8354
Eligible_vs_HighRisk_Precision	0.8517
Eligible_vs_HighRisk_Recall	0.7417
Eligible_vs_HighRisk_F1	0.7929
n_eligible	236.0
n_review	1362.0
n_highrisk	402.0

Table 6 — Multi-seed Stability (5 seeds, XGBoost vs LR) (Synthetic data)

Seed	XGBoost AUC	LR AUC
42	0.7234	0.6977
123	0.7333	0.7158
456	0.7169	0.6784
789	0.7394	0.6828
999	0.7187	0.6924
Mean $\pm$ Std	0.7263 $\pm$ 0.0087	0.6934 $\pm$ 0.0131

Table 7 — External Reference Comparison (external results from different real-world data)

Reference	AUC	Brier	Note
Yadav et al. (individual)	0.765	0.195	External, different data
Yadav et al. (MSME)	0.784	0.184	External, different data
Ots et al.	0.62	nan	External, different data
Ng et al.	0.806	nan	External, different data
Ours (XGBoost, synthetic)	0.7214	0.2072	Synthetic data only
Ours (LR, synthetic)	0.6977	0.2162	Synthetic data only
Ours (RF, synthetic)	0.6993	0.2198	Synthetic data only

Table 8 — XGBoost Best Hyperparameters (CV AUC=0.7352) (Synthetic data)

Hyperparameter	Value
subsample	0.6
reg_lambda	2.0
reg_alpha	0.5
n_estimators	400
min_child_weight	1
max_depth	7
learning_rate	0.01
gamma	0.2
colsample_bytree	1.0

Section 4

Figures

Figure 1 — ROC Curves (Synthetic Data)

ROC Curves — All Models
(Synthetic data — methodological feasibility only)

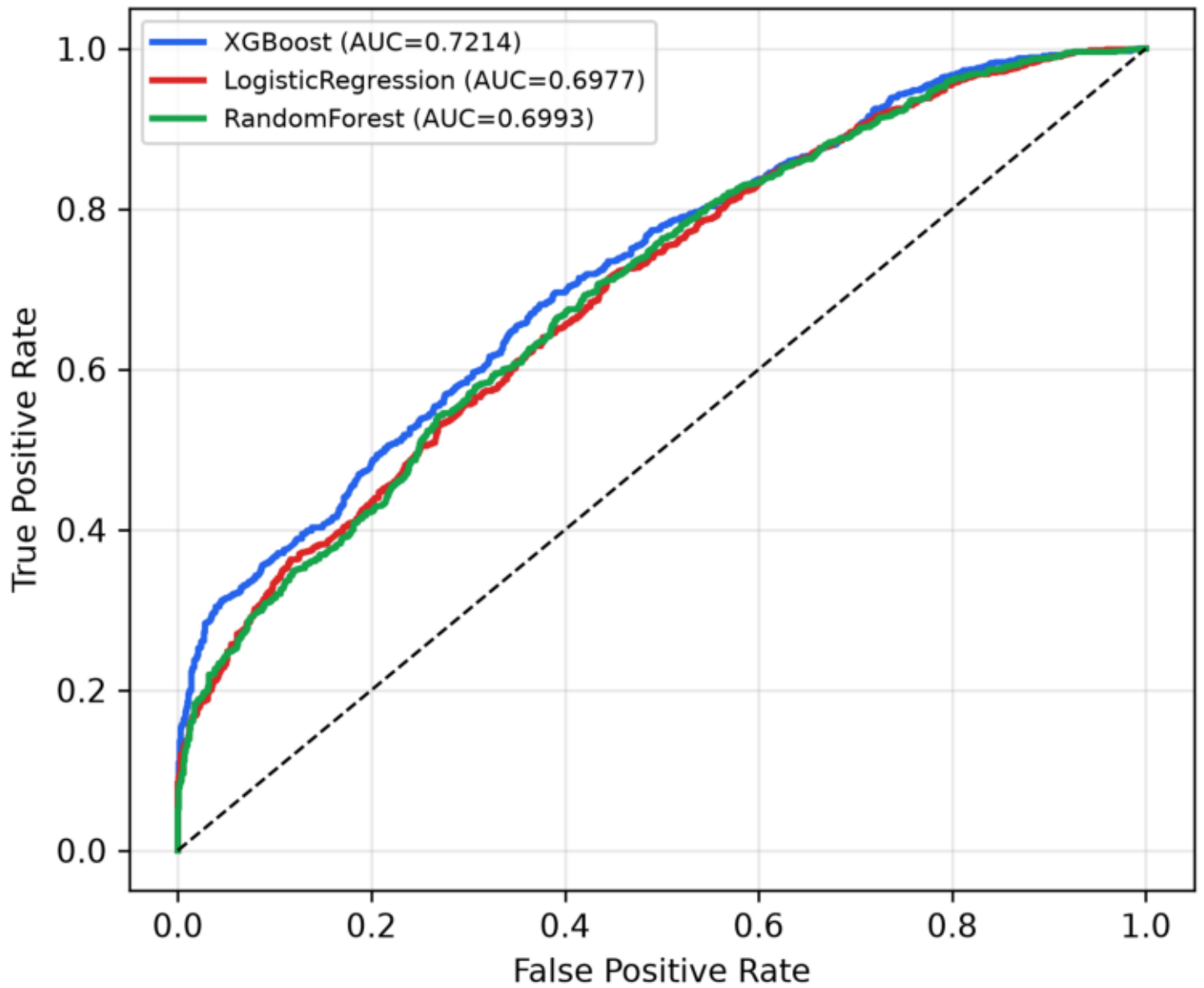


Figure 1 — ROC curves for XGBoost, Logistic Regression, and Random Forest. Synthetic data — methodological feasibility only.

Figure 2 — Score Distribution by True Class (Synthetic Data)

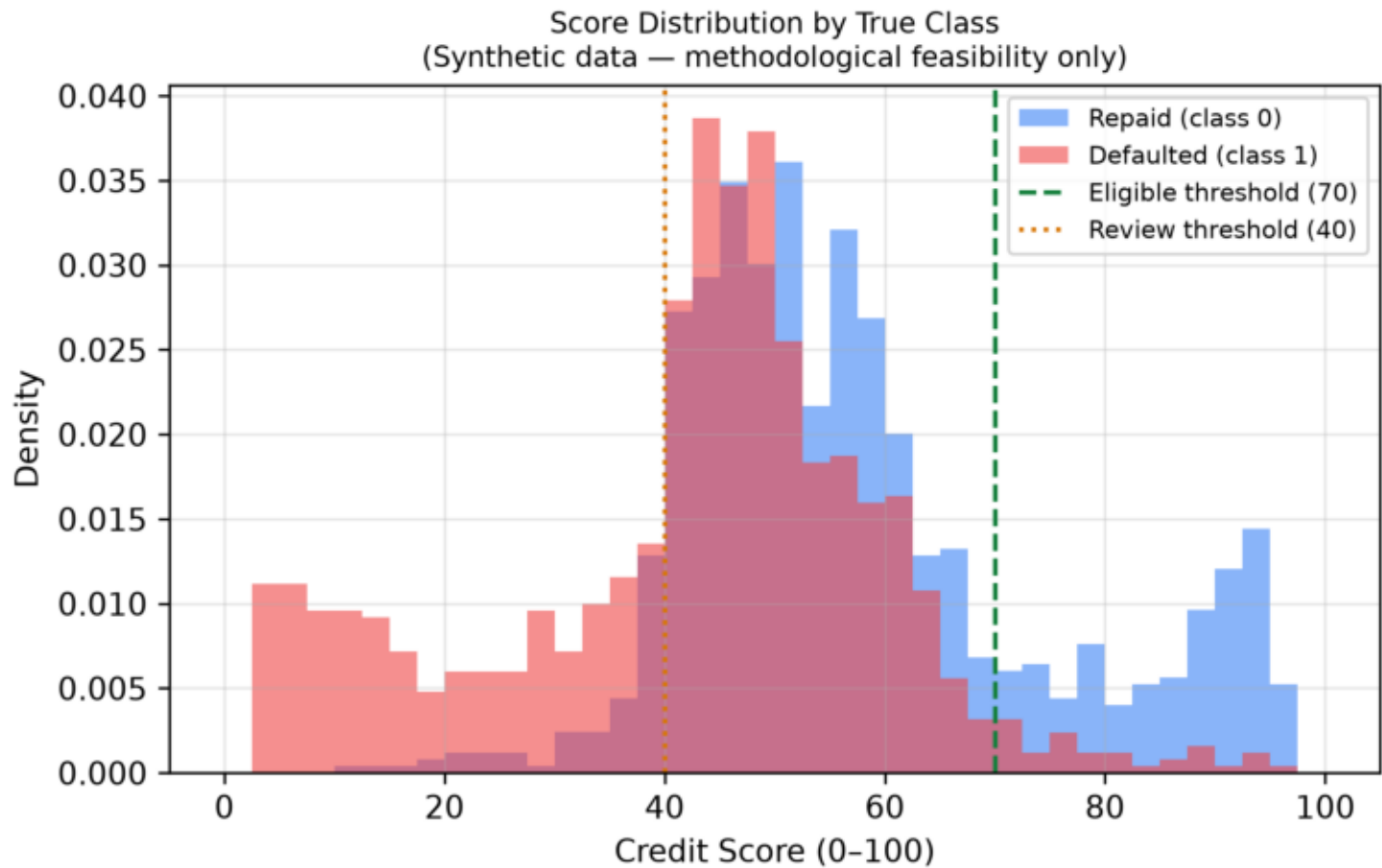


Figure 2 — Distribution of credit scores (0-100) by true class. Dashed lines indicate tier boundaries. Synthetic data — methodological feasibility only.

Figure 3 — Calibration Curve (Synthetic Data)

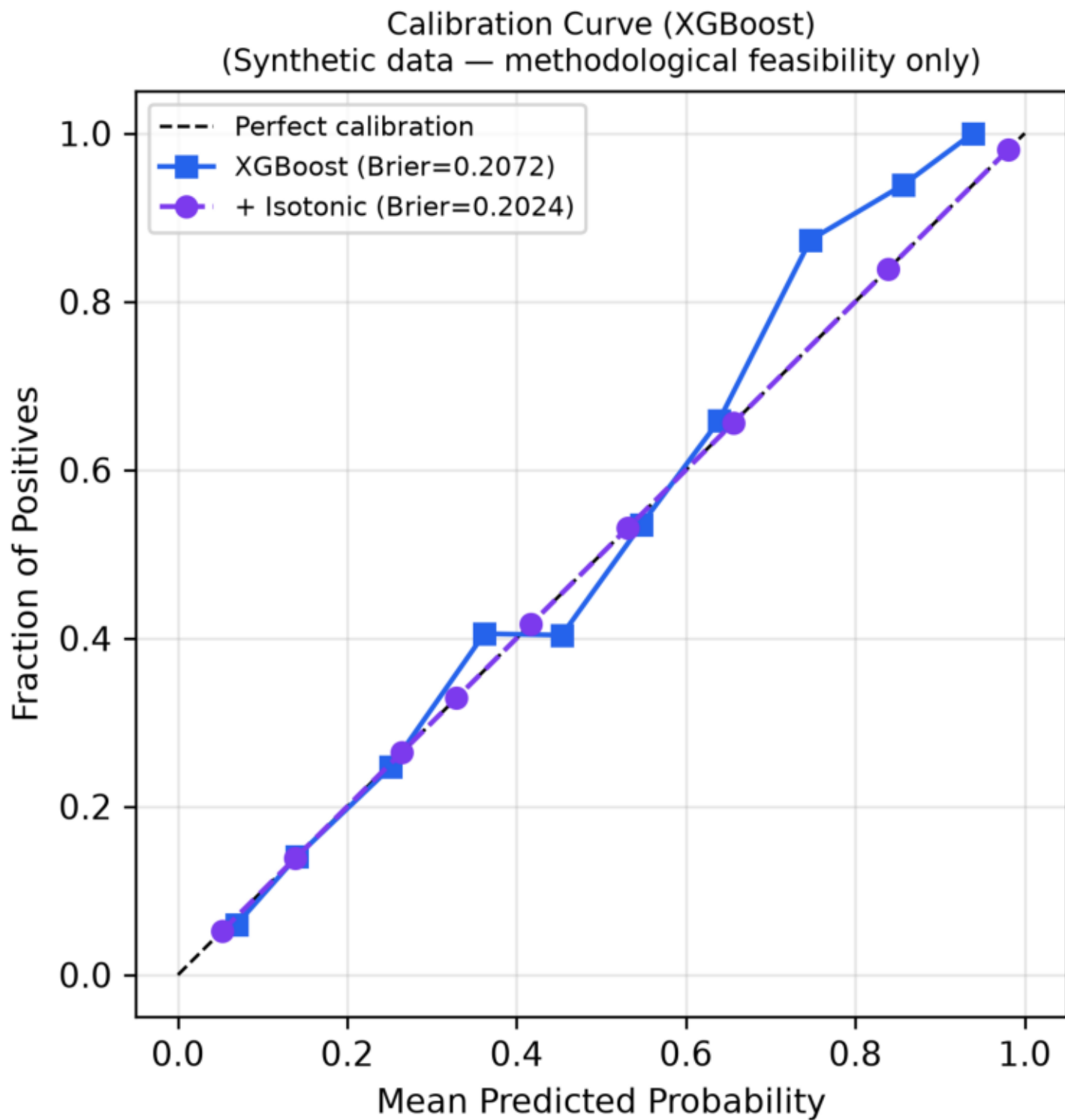


Figure 3 — Reliability (calibration) curve for XGBoost, before and after isotonic regression. Synthetic data — methodological feasibility only.

Section 5

Ablation and Cold-start Results

Table 9 — Ablation: XGBoost with UPI-only, Recharge-only, Combined (Synthetic data)

Feature Set	AUC	Accuracy	Precision	Recall	F1	Brier	cv_auc
UPI_only	0.7124	0.625	0.617	0.665	0.6401	0.2077	0.7223
Recharge_only	0.5447	0.5345	0.5349	0.5503	0.5425	0.2485	0.5559
Combined	0.7234	0.649	0.64	0.6859	0.6622	0.2066	0.7351

Figure 4 — Ablation Study (Synthetic Data)

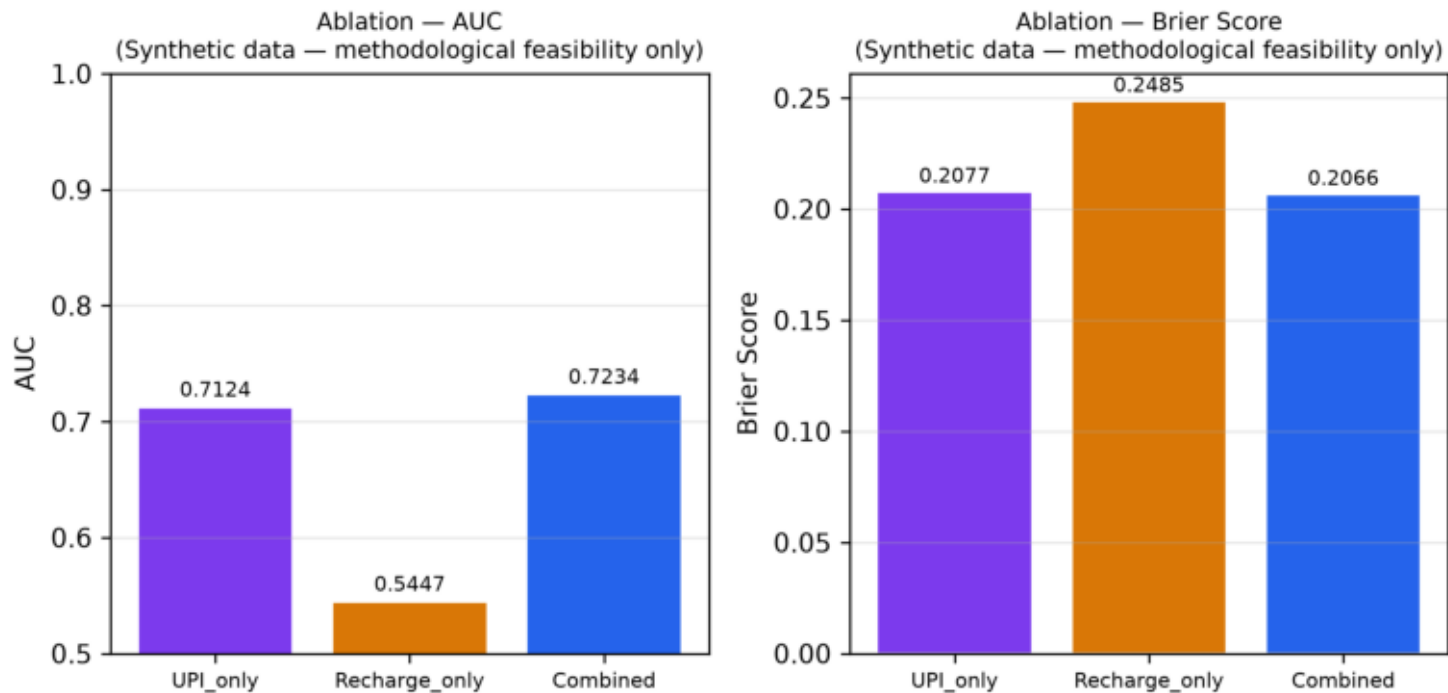


Figure 4 — Ablation study: AUC and Brier score for UPI-only, Recharge-only, and Combined features. Synthetic data — methodological feasibility only.

Figure 5 — Tier Analysis (Synthetic Data)

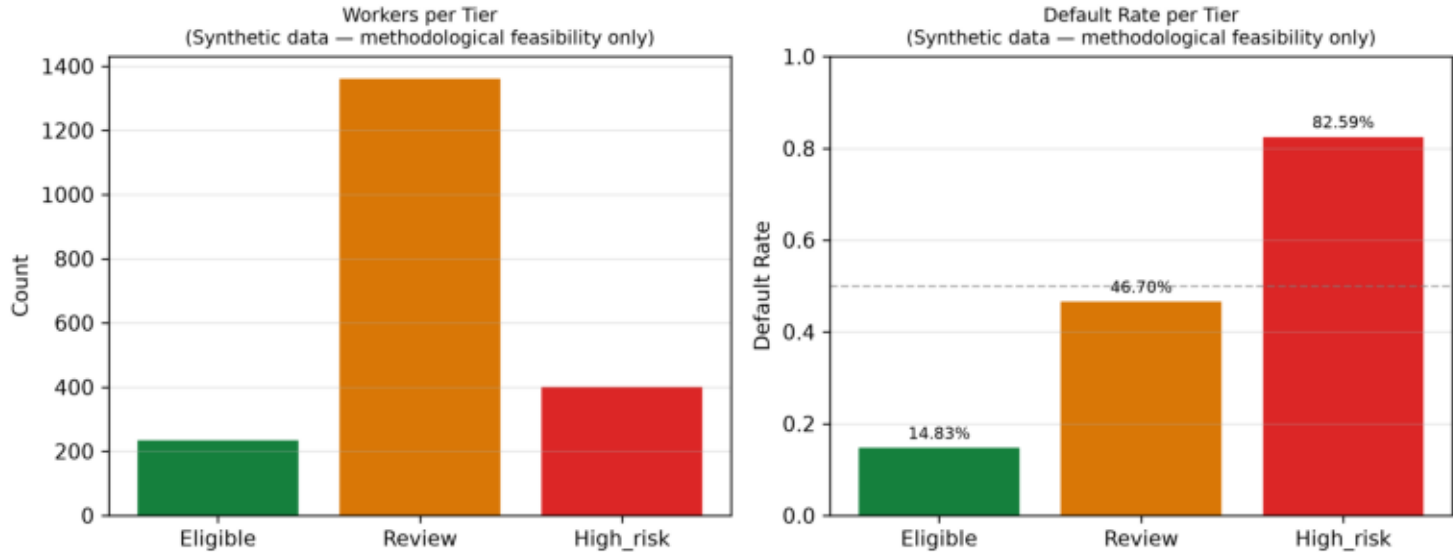


Figure 5 — Worker count and default rate per score tier. Synthetic data — methodological feasibility only.

Table 10 — Cold-start Analysis: Workers with <15 UPI Transactions (Synthetic data)

Metric	Value
n_cold_workers	1042
UPI_only_AUC	0.5126
Combined_AUC	0.5468
UPI_only_Brier	0.2508
Combined_Brier	0.2493

Cold-start and Imbalanced-data Results (Synthetic)

Cold-start test: workers with fewer than 15 UPI transactions in the 90-day window.

- Number of cold-start workers in test set: 1042
- UPI-only AUC on cold-start workers: 0.5126
- Combined AUC on cold-start workers: 0.5468
- UPI-only Brier on cold-start workers: 0.2508
- Combined Brier on cold-start workers: 0.2493

Interpretation: If Combined AUC > UPI-only AUC on cold-start workers, this supports the paper's claim that recharge data provides meaningful signal for workers with sparse UPI histories. The actual direction is reported as computed.

Imbalanced-data Experiment (80/20 ratio):

- Workers: 5000
- Positive rate: 0.23
- XGBoost AUC: 0.677
- XGBoost Brier: 0.1519

Figure 6 — Multi-seed Stability (Synthetic Data)

Multi-seed Stability (5 seeds)
(Synthetic data — methodological feasibility only)

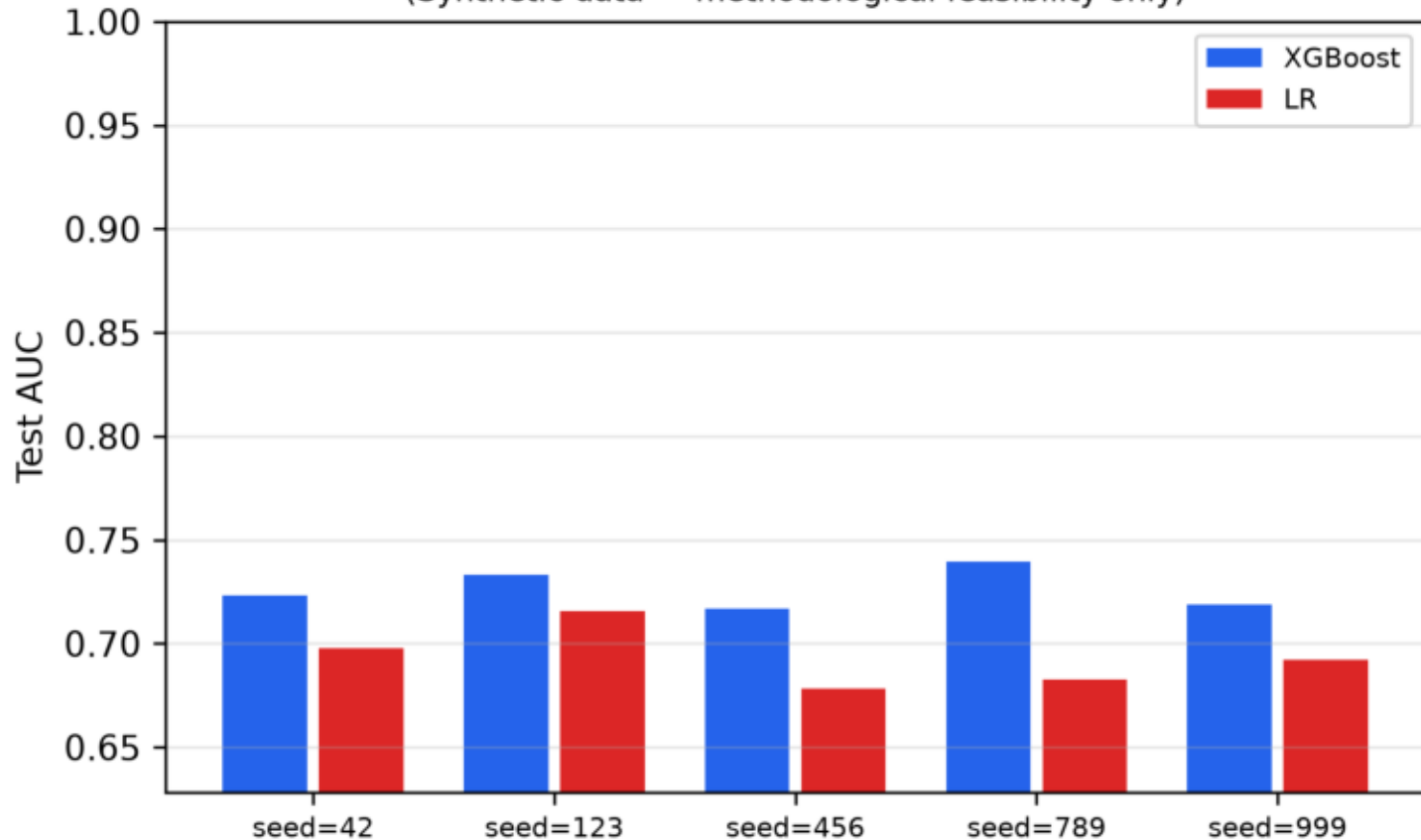


Figure 6 — Multi-seed stability: XGBoost and LR test AUC across 5 different random seeds. Synthetic data — methodological feasibility only.

Section 6

Fairness Analysis

Table 11 — Fairness: Eligible-tier Approval Rates by Group (Synthetic data)

Group Type	Group	Approval Rate
Income Tercile	Low	0.0000
Income Tercile	Mid	0.0000
Income Tercile	High	0.3538
Worker Category	daily_wage_labourer	0.0084
Worker Category	domestic_worker	0.0000
Worker Category	gig_delivery	0.3616
Worker Category	street_vendor	0.0886

Fairness Interpretation (Synthetic Data)

Disparate Impact Ratio (DIR) = $\min(\text{group_rate}) / \max(\text{group_rate})$

Acceptable range per guidelines: 0.80 – 1.25

DIR by Income Tercile : 0.0

DIR by Worker Category : 0.0

Note: A DIR < 0.80 suggests potential adverse impact on a disadvantaged subgroup.

A DIR > 1.25 suggests preferential treatment. Actual values are reported as computed from the synthetic simulation without post-hoc adjustment.

Limitations: In a real deployment, additional protected attributes (gender, caste, region) and intersectional analysis would be required. The synthetic data does not encode protected attributes beyond worker category and income.

Figure 7 — Fairness Analysis (Synthetic Data)

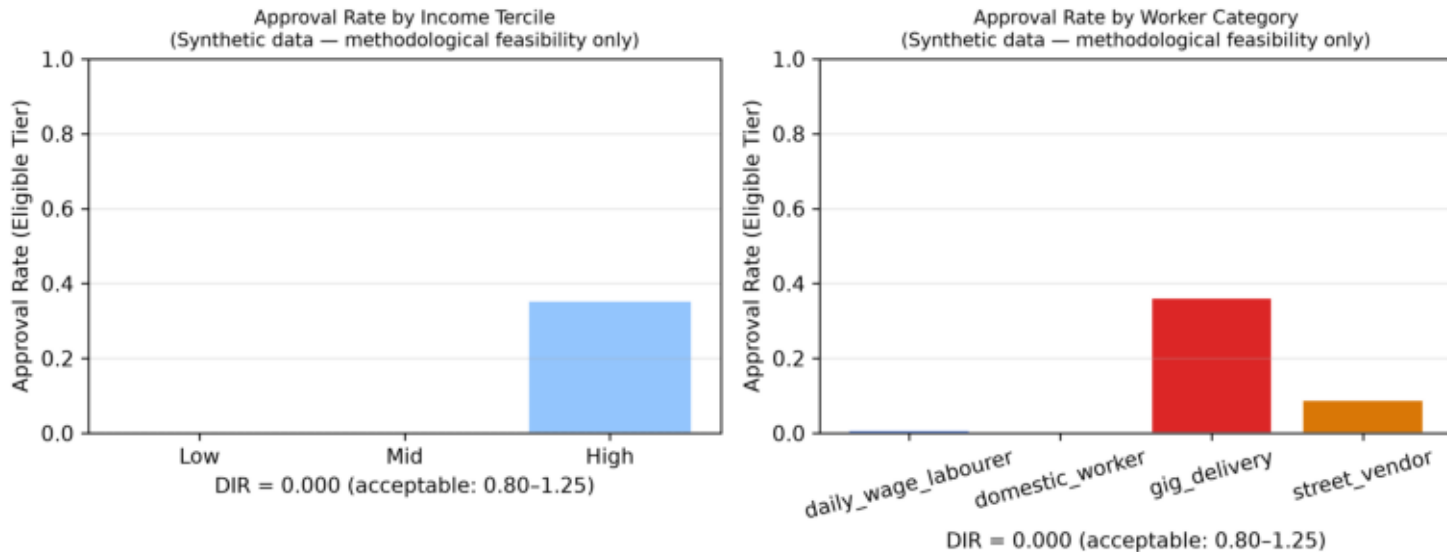


Figure 7 — Approval rates (Eligible tier) by income tercile and worker category. Synthetic data — methodological feasibility only.

Section 7

SHAP Explainability

Figure 8 — SHAP Global Bar Plot (Synthetic Data)

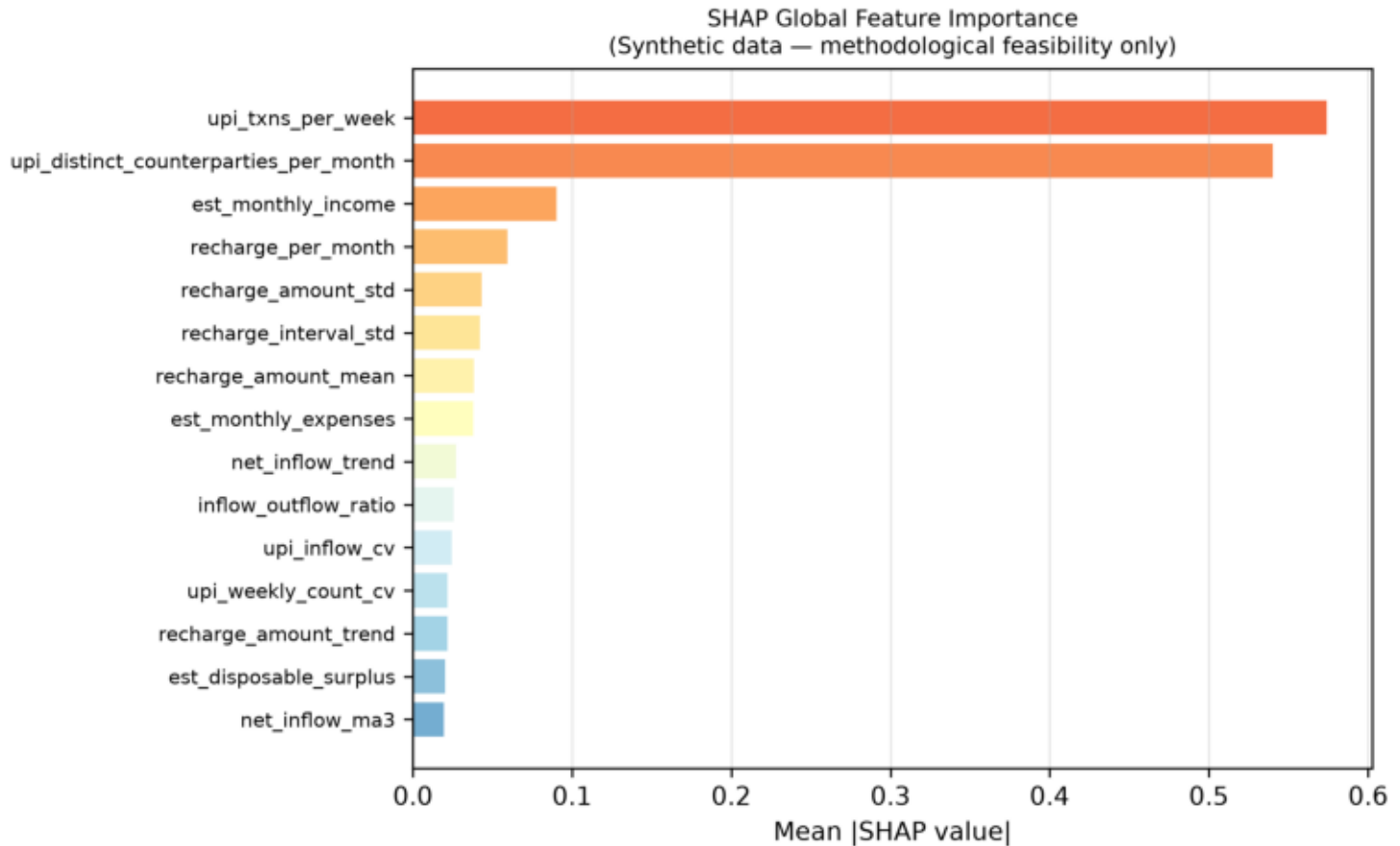


Figure 8 — Global SHAP feature importance: mean |SHAP value| across test sample. Synthetic data — methodological feasibility only.

Figure 9 — SHAP Beeswarm Summary (Synthetic Data)

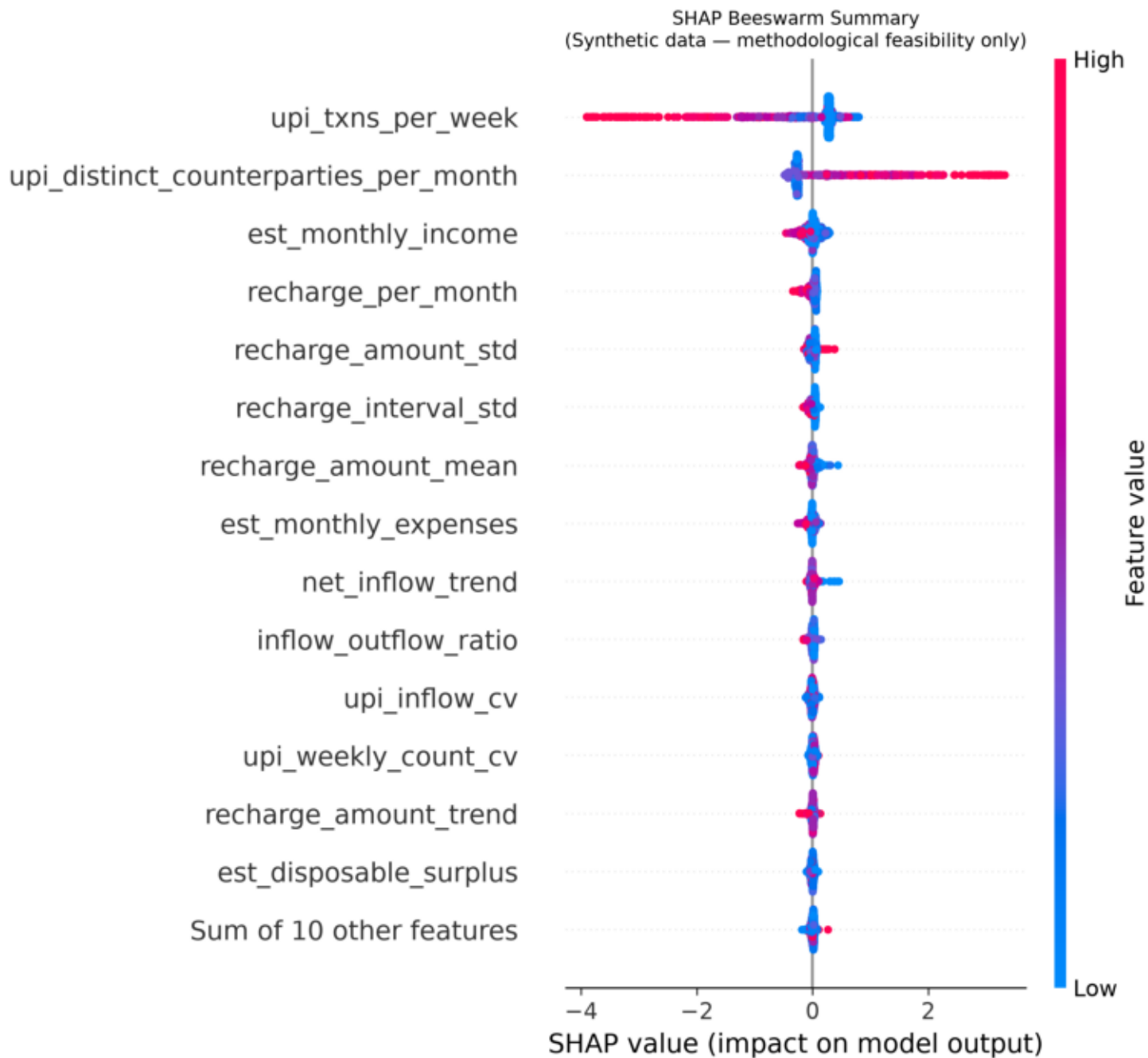


Figure 9 — SHAP beeswarm summary: each point is a worker, colour indicates feature value. Synthetic data — methodological feasibility only.

Individual SHAP Explanations — Synthetic Data

SHAP Individual Explanations (scaled to score contribution points)

Formula: $\text{score_pts} \approx \text{SHAP_value} \times (-100 \times p \times (1-p))$

Sanity check: $\text{sum}(\text{SHAP values}) \approx \text{model_output} - \text{base_value}$

--- Eligible Worker ---

Credit Score : 79.6 / 100

P(default) : 0.2037

SHAP base val: 0.0033

Sum(SHAP) : -1.3665 [sanity: diff from base = -1.3665]

Top driving features:

- +20.1 pts ← Weekly UPI transaction frequency = 15.08 (SHAP=-1.2406)
- 7.6 pts ← Distinct counterparties per month = 6.00 (SHAP=+0.4710)
- +3.9 pts ← Estimated monthly expenses (UPI debits, INR) = 16579.83 (SHAP=-0.2374)
- +3.3 pts ← Estimated monthly income (UPI credits, INR) = 19428.13 (SHAP=-0.2021)
- +1.5 pts ← Mean recharge plan amount (INR) = 199.00 (SHAP=-0.0901)

--- Review Worker ---

Credit Score : 44.7 / 100

P(default) : 0.5527

SHAP base val: 0.0033

Sum(SHAP) : 0.2084 [sanity: diff from base = 0.2084]

Top driving features:

- 6.7 pts ← Weekly UPI transaction frequency = 0.08 (SHAP=+0.2694)
- +6.4 pts ← Distinct counterparties per month = 0.33 (SHAP=-0.2600)
- +2.6 pts ← Estimated monthly income (UPI credits, INR) = 85.06 (SHAP=-0.1052)
- 2.4 pts ← Mean recharge plan amount (INR) = 0.00 (SHAP=+0.0955)
- 1.7 pts ← Mobile recharge frequency per month = 0.00 (SHAP=+0.0698)

--- High_risk Worker ---

Credit Score : 32.2 / 100

P(default) : 0.6779

SHAP base val: 0.0033

Section 8

Operational Latency

Section 8 — Operational Latency

Single-worker prediction + SHAP explanation latency (200 repetitions, in-process):

Mean latency : 11.21 ms

P95 latency : 11.73 ms

Measurement method: Python `time.perf_counter()` around `predict_proba()` + `TreeExplainer()` for a single-row input. No network overhead. Measured on the development machine.

Hardware: laptop CPU (x86_64), no GPU.

These latency figures are suitable for real-time loan application scoring, where sub-second response is typically required.

Section 9

Limitations and Threats to Validity

Section 9 — Limitations and Threats to Validity

⚠ CRITICAL DISCLAIMER ⚠

ALL DATA IN THIS STUDY IS ENTIRELY SYNTHETIC. No real UPI transaction records, mobile recharge histories, or loan-repayment outcomes were used. The synthetic generator is calibrated to approximate realistic statistical properties of informal workers in India, but it cannot capture the full complexity of real-world data.

RESULTS INDICATE METHODOLOGICAL FEASIBILITY ONLY — NOT REAL-WORLD PERFORMANCE.

SPECIFIC THREATS TO VALIDITY:

1. Synthetic data bias: The generator's distributional assumptions may not match the true joint distribution of UPI behaviour and creditworthiness. Real data may reveal different feature importance rankings.
2. Label noise simulation: ~6% label flip is an approximation. Real default labels may have higher or structured noise (e.g., reporting lag).
3. No temporal leakage check: In a real deployment, strict point-in-time splits would be required to prevent look-ahead bias.
4. Synthetic feature correlations: The generator creates correlations through the latent capacity variable, but real-world correlations may differ substantially in magnitude and direction.
5. Fairness analysis scope: Only income tercile and worker category are analysed. Real-world fairness audits require protected attributes (gender, geography, caste, religion) and intersectional analysis.
6. Regulatory compliance: Credit-scoring models in India are subject to RBI/NBFC regulations. A production system would require extensive regulatory review independent of this simulation.
7. Feature engineering assumptions: EMI detection, income estimation,